

Pratik Patil

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Backend **Software Engineer** with **3+ years** building production backend systems in **Java (Spring Boot)**. Architected the Central Server Manager, used by **600+ engineers** – improving provisioning speed by **33%** and cutting API response times from seconds to milliseconds via Redis caching and query optimization. Also builds and ships the CI/CD pipelines and cloud infrastructure behind it.

Experience

Software Engineer	PTC Software (India) Pvt. Ltd.	<i>Jul 2024 – Present</i>
- Architected and led development of the Central Server Manager (CSM), a Spring Boot-based platform used by 600+ engineers, improving server provisioning speed by approx. 33% during testing phases. - Implemented dynamic, server-side filtering using Spring Data JPA Specifications and pagination across a 10,000+ row dataset – cutting multi-second load times to sub-second responses. - Optimized a high-volume data-fetch API – combining Redis caching, query indexing, and server-side filtering – reducing response size from 22MB to 277KB and latency from 7s to 225ms. - Diagnosed and fixed a fork-from-thread deadlock in Ansible playbook execution; re-architected long-running automation using Spring Boot @Async with dedicated TaskExecutor worker pools, eliminating silent job hangs in production. - Designed RBAC with Spring Security, integrating LDAP/Active Directory authentication and JWT-based role authorization across CSM. - Engineered a GitLab CI/CD pipeline (build, test, package, deploy) with JUnit/Mockito test stages, zero-downtime rolling deployments to Kubernetes (AKS), and automated secret injection via Azure Key Vault. - Automated Azure infrastructure cleanup via Python and Azure CLI – VM auto-expiry tagging, weekend deallocation, and removal of orphaned NICs/disks – reducing unnecessary cloud spend. - Improved system observability via global exception handling and SLF4J structured logging, cutting log-based issue diagnosis time from 30 minutes to 10-15 minutes. - Owned the UBER base image Jenkins pipeline for Windchill installations, maintaining multi-stage build logic to support new releases with automated failure alerts.		
Software Engineer Intern	PTC Software (India) Pvt. Ltd.	<i>Jul 2023 – Jun 2024</i>
- Built Testharmonics (Django/React) from a bare skeleton into a self-service tool for triggering and tracking Jenkins jobs, logging trigger history, usage duration, and job metadata to PostgreSQL, with a UI surfacing job status, stages, and logs. - Standardized mail notification formatting across all Jenkins jobs in Testharmonics – consistent templates for success, failure, and abort alerts, improving clarity of automated status emails. - Ran a POC integrating Backstage (Spotify's developer portal) with existing Jenkins pipelines and Confluence documentation, centralizing service and build visibility into a single internal portal.		

Technical Skills

Languages	Java, Python, TypeScript, JavaScript, C++, SQL
Java	Java 17, Multithreading & Concurrency, Streams & Lambda
Backend & APIs	Spring Boot, Hibernate, JPA, REST APIs, Microservices, Spring Security, JWT, OAuth2, Django
Frontend	React, Angular, Redux, Context API, HTML, CSS
Databases	PostgreSQL, MongoDB, MySQL
Caching & Messaging	Redis, Event-Driven Architecture, RabbitMQ
DevOps & Cloud	Azure, Docker, Kubernetes, Jenkins, GitLab, CI/CD, Maven, Terraform, Grafana, Ansible, Shell Scripting
Testing & Quality	JUnit, Mockito, Postman, Unit & API Testing
Core Concepts	System Design, Distributed Systems, DSA, OOP, DBMS, OS, Computer Networks, Agile/Scrum

Projects

LinkZen Spring Boot, React, MongoDB, Redis, RabbitMQ, Docker	<i>2026</i>
- Architected a URL shortener with sub-10ms redirect latency via Redis caching; MongoDB schema uses TTL indexes for auto-expiry and compound indexes for analytics queries. - Designed an async click analytics pipeline using RabbitMQ—redirect requests publish click events asynchronously to a queue; a consumer processes analytics with a 3-retry DLQ fallback, decoupling database writes from the critical path. - Implemented Redis-based rate limiting per user/IP and JWT-based stateless authentication; containerized full stack with Docker for reproducible deployment.	

Education

Bachelor of Engineering in Computer Engineering	<i>2020 – 2024</i>
D.Y. Patil College of Engineering, Pune	CGPA: 9.0 / 10.0

Achievements

- Flipkart GRID 4.0:** Ranked Top 5% **nationwide** out of thousands of applicants in one of India's largest engineering challenges.
- Competitive Programming:** LeetCode **460+** · CodeChef **1680** · Codeforces **1260** · **500+** DSA problems solved across platforms.